

Telecom Customer Churn Analysis

Strategic Executive Report & Business Insights

20.3%

Average Churn Rate

\$5,350.60

Monthly Revenue at Risk

21.8%

% Total Revenue Impact

1. Executive Summary

This analysis investigates the primary drivers of customer attrition within the telecom portfolio. With a churn rate of **20.3%**, the business faces a significant financial impact, losing over **\$5,350.60** in monthly recurring revenue. The primary goal of this EDA was to transition from reactive data monitoring to proactive retention strategies.

2. Key Research Findings

A. The "Point of No Return" in Customer Service

Analysis reveals a critical friction point at **4 customer support calls**. At this threshold, the probability of churn spikes exponentially from ~24% to over 45%. This suggests a systemic failure in resolving technical or billing issues within the first three attempts.

B. Geographic & Infrastructure Impact

Network Quality is the primary retention driver in Urban areas. Customers experiencing "Poor" quality scores in cities churn at nearly triple the rate of those with "Good" quality scores. In contrast, Suburban churn is more closely tied to plan type and pricing.

C. High-Value Segment Vulnerability

Contrary to common assumptions, churn is not limited to low-value or "budget" plans. The **Family Plan** segment exhibits the highest churn rate (27%), indicating that the loss of a single account often represents the loss of multiple household lines.

3. Strategic Recommendations (Recruiter-Ready)

- ✓ **Automated Retention Triggers:** Implement an immediate "High-Priority Save" protocol for any customer reaching their 3rd support call. Offering a loyalty discount or a technical health check at this stage could prevent the 4-call churn spike.
- ✓ **Infrastructure ROI Prioritization:** Technical upgrades should be prioritized in Urban high-churn zones. Data shows that improving the Network Quality Score from 2 to 4 can reduce churn by nearly 60% in these segments.
- ✓ **Family Plan Loyalty Programs:** Since Family Plans are high-revenue and high-churn, introduce a "Length of Loyalty" bonus for multi-line accounts to increase switching costs.
- ✓ **Transition to Auto-Debit:** Incentivize customers to move away from E-wallet/ Cash payments, as friction in the monthly payment process was correlated with higher churn.

4. Conclusion

By leveraging data-driven insights, the business can shift from a "wait-and-see" approach to a targeted, predictive model. The identified KPIs—specifically the Support Call Threshold and Network Quality Scores—provide a clear roadmap for reducing the 21.8% revenue risk and stabilizing the long-term customer base.